

RECURSIVE BIOFEEDBACK MODEL OF THE SELF (MBRS)

Scientific White Paper — Version 3.0

A mechanistic hypothesis for the emergence of Self from recurrent appropriation, referential continuity and organism–world biofeedback

CENTRAL CLAIM

The MBRS proposes that the Self need not pre-exist as the owner of experience. A sufficiently coherent and recurrent organization of belonging relations — bodily, experiential, mnemonic, causal and temporal — may generate a stable referential invariant that is experienced and functionally used as «I». Once constituted, the Self recursively modulates valuation, attention, prediction and action, and can extend its domain of belonging beyond the biological body.

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Conceptual and computational research developed in dialogue with ChatGPT (OpenAI)

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Abstract

The human Self is ordinarily experienced as a unitary owner of a body, experiences, memories and actions, yet neuropsychological and experimental evidence indicates that body ownership, self-location, agency, autobiographical memory and source monitoring are at least partly dissociable. This scientific white paper presents the Recursive Biofeedback Model of the Self (MBRS), a mechanistic framework in which Self is not assumed as a pre-existing controller but emerges from recurrent relations of belonging organized around an organism-centred reference. The model distinguishes total informational flow (R), non-conscious distributed processing (N), transient integration (G), conscious experiential flow (C), a multidimensional appropriation vector (Ω), referential continuity (ΓR), and the emergent Self (S). Its constitutive hypothesis is that recurrent appropriation precedes and helps generate Self, after which Self and appropriation enter a bidirectional recursive loop. Exploratory simulations show that a functional Self-like architecture can be constructed without a phenomenal variable; that a causally inert phenomenal placeholder is empirically invisible; that plastic memory introduces historical dependence; and that selective ablation of ownership dimensions yields dissociation profiles. These profiles are compared with published findings on disturbed limb ownership, agency/ownership dissociation, autobiographical amnesia, source-monitoring biases, dissociative identity disorder and depersonalization. The available evidence is compatible with, but does not validate, the proposed decomposition. The paper therefore defines differential predictions, competing models, and explicit falsification conditions. MBRS is presented as a theory of Self formation and dynamics, not as a solution to the phenomenal problem of consciousness.

Keywords: Self; bodily self-consciousness; ownership; appropriation; agency; autobiographical memory; source monitoring; recursive biofeedback; interoception; consciousness.

Scientific status and scope

EPISTEMIC STATUS

MBRS v3.0 is a falsifiable theoretical framework under development. It is not a validated clinical model, does not infer consciousness from biological feedback alone, and does not claim that computational simulations instantiate subjective experience. Compatibility with existing data is treated as constraint, not confirmation.

The primary explanandum is the emergence and persistence of a Self once experience is present. The separate question of why organized biological activity is accompanied by phenomenal experience remains open. Contemporary consciousness science itself remains theoretically pluralistic; recent reviews and adversarial tests continue to compare Global Neuronal Workspace, Integrated Information, Recurrent Processing, Predictive Processing and related frameworks without convergence on a single accepted theory [1]–[4].

Research questions

- Can a Self be modelled without positing a central observer or executive homunculus?
- Are relations of belonging/appropriation constitutive of Self, rather than merely consequences of an already existing Self?
- Can body ownership, experiential belonging, memory ownership, action authorship and source attribution dissociate while a minimal present Self persists?
- Does referential continuity across changing states provide a better invariant of Self than any stable content?
- What observations would falsify the appropriation hypothesis?

1. Introduction

Everyday experience suggests a deceptively simple architecture: a subject receives perceptions, owns a body, remembers a past, chooses actions and observes their consequences. Neuroscience makes that picture difficult to sustain. Perception, interoception, memory, valuation, motor control and prediction are distributed processes; bodily self-consciousness itself can be decomposed into partially separable components such as body ownership,

self-location and first-person perspective [5]. Recent neuroimaging meta-analysis likewise reports both distinct and overlapping neural patterns for manipulated body ownership and sense of agency, emphasizing multisensory integration and self-attribution rather than a single anatomical Self centre [6].

MBRS begins from this distributed architecture and asks a narrower question than a general theory of consciousness: given an organism with an experiential stream, how can successive states become organized as belonging to the same being? The proposed answer is not a stored Self-object but a recurrent relation. Experiences, bodily states, memories and actions repeatedly converge on the same organism-centred reference; plasticity and memory preserve the consequences of that convergence; referential continuity links successive configurations; the resulting relational invariant is what the model calls Self.

$$\text{LIFE} \neq \text{CONSCIOUSNESS} \neq \text{SELF}$$

$$\text{Experience} + \text{recurrent belonging} + \text{referential continuity} \rightarrow \text{emergent Self}$$

1.1 Contribution claimed — and not claimed

The proposed contribution is an explicit causal architecture for Self formation centred on recurrent appropriation and referential continuity. Individual ingredients — interoception, bodily ownership, agency, predictive processing, autobiographical memory, recurrent integration and self-models — are not new. Novelty, if established, must reside in the organization of these elements, the directionality claim appropriation → Self, and the differential predictions generated by the model. A systematic novelty review remains necessary before any priority claim.

2. Constructs and operational definitions

Symbol / construct	Operational role
R(t) — informational river	Continuously changing internal and external signals available to the organism; larger than conscious content.
N(t) — non-conscious processing	Distributed construction of maps, salience, causal attribution, predictions and candidate contents.
G(t) — transient integration	Temporarily coordinated configuration selected from distributed processing; functional construct, not equated with consciousness.
C(t) — conscious stream	Temporal experiential flow. Its phenomenal existence is assumed as explanandum boundary, not explained by fiat.
F(t) — focus	Momentarily privileged subset of conscious content; F(t) is not the Self.
P(t) — proto-Self	Pre-reflective organism-centred reference integrating bodily state, position and action-related signals.
Ω(t) — appropriation vector	Multidimensional relations of belonging/self-attribution.
ΓR(t) — referential continuity	Persistence of the relation «this same being» across changing states.
S(t) — Self	Emergent virtual, temporal and relational organization that supports present self-reference and recursively modulates later states.
ΩE — extensive appropriation	Post-emergence extension of belonging to objects, places, relationships, projects, symbols and social identities.

2.1 Appropriation is not possession

In MBRS, appropriation is a technical term for a relation of belonging or self-attribution. At the constitutive level it concerns body, experience and action; at the extensive level it may concern objects, relationships or symbols. The model does not imply legal or moral ownership of other persons. Self–Other differentiation is an independent constraint.

3. Core architecture

$$R(t) \rightarrow N(t) \rightarrow G(t) \rightarrow C(t) \rightarrow \Omega(t) \rightarrow \Gamma R(t) \rightarrow S(t)$$

$$S(t) \rightarrow \{V(t+1), F(t+1), \text{Pred}(t+1), A(t)\} \rightarrow \text{body/world}(t+1) \rightarrow R(t+1)$$

The arrows are schematic. The biological system contains recurrent and bidirectional couplings at every level. The purpose of the diagram is to distinguish constructs that are often collapsed into one another: incoming information, non-conscious construction, conscious experience, belonging, temporal continuity and Self.

3.1 The orchestra without a conductor

$$G(t) = \mathcal{G}[P_1(t), P_2(t), \dots, P_n(t), W(t)]$$

No process P_{maestro} is postulated. Vision, audition, interoception, proprioception, vestibular processing, memory, threat, reward, language, prediction and motor control operate in parallel and recurrently constrain one another. Global availability and recurrent amplification are prominent in workspace accounts of conscious processing [2], [7], while recurrent-processing approaches emphasize recurrent interactions as critical to awareness [8]. MBRS borrows neither as a sufficient explanation of phenomenality; it uses distributed recurrence as architectural background.

3.2 The river and the focus

$$R(t) \neq C(t) \neq F(t) \neq S(t)$$

The informational river continues independently of voluntary focus. Non-conscious processing does more than filter; it constructs spatial, bodily, affective and causal organization before report. Consequently, conscious content may already arrive as body-centred or action-related. This opens the possibility that a pre-personal form of belonging begins before explicit self-recognition.

4. Biological grounding: regulation, interoception and history

$$x(t+1) = F[x(t), e(t), v(t), r(t), m(t)]$$

Living organisms continuously regulate internal variables and act on the environment. MBRS treats this recursive regulation as necessary background, not as sufficient evidence of consciousness. Interoceptive signals interact with exteroceptive and multisensory information and are increasingly studied as contributors to bodily self-awareness and Self–Other differentiation [9].

$$W(t+1) = W(t) + \Delta W[E(t), V(t), A(t), M(t)]$$

Plasticity turns regulation into regulation-with-history. The same current bodily state need not yield the same next state if previous experience has modified weights, priors or memory. This historical dependence is essential to MBRS because the Self is a trajectory, not an instantaneous snapshot.

4.1 Proto-Self

$$P(t) = P[B(t), \text{interoception, proprioception, vestibular state, nociception, action consequences}]$$

The proto-Self is deliberately pre-reflective. It is not a miniature conscious ego. It denotes the organism-centred reference that makes distinctions such as here/not-here, body/world and consequences-of-this-organism possible. This terminology is broadly compatible with body-mapping traditions but MBRS does not assume that proto-Self alone explains feeling.

5. Consciousness and the phenomenal boundary

MBRS separates its Self hypothesis from the hard phenomenal question. A system can integrate signals, monitor itself, remember, predict, report and act without the model thereby establishing that there is something it is like to be that system. Contemporary theory comparisons remain divided over what mechanisms are necessary or sufficient for consciousness [1], [3]. The 2025 adversarial comparison of GNWT and IIT provided important theory-constraining results but did not collapse the field into a single accepted account [4].

PHENOMENAL BOUNDARY

MBRS v3.0 explains neither why biological processing is accompanied by experience nor whether phenomenality is an additional causal property. It asks: once experience exists, how does an owner-like Self emerge from it?

A future version may test the stronger identity hypothesis that certain organized biological dynamics are themselves experiential, rather than causing an additional phenomenal state. That hypothesis is currently outside the evidential claims of MBRS.

6. The constitutive appropriation hypothesis

Conventional intuition: SELF $\rightarrow$ «this experience is mine»

MBRS hypothesis: recurrent «belongs to this being» $\rightarrow$ SELF

The model's central inversion is ontogenetic and mechanistic. A fully formed Self is not required for pre-reflective body-centred attribution. Recurrent belonging relations can be established around the organism before autobiographical identity or explicit self-concepts exist. If these relations remain coherent through changing content, they can generate an invariant reference.

$$\Omega(t) \rightarrow S(t) \rightarrow \Omega(t+1)$$

Once Self emerges, causality becomes bidirectional. The Self affects what is attended, valued, remembered, defended and incorporated into its domain. Therefore adult observation alone cannot establish whether appropriation originally caused Self or vice versa; developmental and perturbational tests are required.

6.1 Multidimensional Ω

$$\Omega(t) = [\Omega B, \Omega X, \Omega M, \Omega A, \Omega O, \Omega E, \dots]$$

Dimension	Meaning	Primary dissociation target
ΩB — body ownership	Body or body-part represented as belonging to this organism.	Disturbed limb ownership / somatoparaphrenia
ΩX — experiential belonging	Experience situated as happening to/from this organism.	Depersonalization / experiential detachment
ΩM — mnemonic belonging	Memory represented as belonging to this life-history.	Amnesia / dissociative access
ΩA — action authorship	Action represented as caused or initiated by this organism.	Agency disturbances / alien or anarchic action
ΩO — source attribution	Content attributed to internal vs external origin.	Source-monitoring biases / hallucination research
ΩE — extensive belonging	Object, place, relationship, project or symbol integrated into Self-domain.	Self-reference and loss/valuation paradigms

7. Referential continuity and the temporal Self

$$S(t) \neq S(t+1), \text{ yet } S(t) \sim S(t+1) \text{ through } \Gamma R$$

Self is not defined by invariant content. Body state, emotion, thought and accessible memory change continuously. MBRS therefore locates invariance in the reference relation: changing states continue to be attributed to the same being. ΓR is a candidate relational invariant, not a new anatomical module.

7.1 Minimal, autobiographical and reflective Self

Level	Phenomenological shorthand	Dominant requirements
Minimal Self	I — here — now	$\Omega B, \Omega X$, short-range ΓR
Autobiographical Self	I — through my history	ΩM , memory, long-range ΓR , prediction
Reflective Self	I can represent myself as an object of	language, metacognition, abstraction,

	thought	autobiographical structure
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Neuropsychological amnesia provides an important constraint. A published case showed coherent and continuous self-concept despite inaccessible episodic memories for a period, supported by conceptual autobiographical knowledge [10]. Other medial temporal lobe cases show that autobiographical fact and trait knowledge can also be substantially impaired, demonstrating that memory–Self relations are heterogeneous rather than cleanly modular [11].

8. Action: organismal action, voluntary control and authorship

ORGANISMAL ACTION $\neq$ VOLUNTARY CONTROL $\neq$ FELT AUTHORSHIP

MBRS treats the Self as a participant in closed-loop action, not as a passive spectator. Nevertheless, the existence of movement, voluntary control and the feeling that «I did this» are separable. Experimental and neuroimaging literatures distinguish body ownership and sense of agency, with partially distinct as well as shared neural correlates [6], [12].

$$S(t) \rightarrow A(t) \rightarrow \text{body/world}(t+1) \rightarrow E(t+1) \rightarrow S(t+1)$$

Action adds a second primitive relation to «this happens to this organism»: «this organism makes this happen». The conjunction of perceptual belonging and causal participation is hypothesized to stabilize the minimal Self.

9. Extension of Self beyond the body

constitutive $\Omega \rightarrow \text{SELF} \rightarrow \text{extensive } \Omega E \rightarrow \text{enriched SELF}$

Once a Self exists, belonging can extend beyond biological boundaries. Personal objects, places, relationships, projects, reputation, social groups and ideas can become Self-linked. This does not make them biological parts of the organism; it means their representation acquires Self-referential value.

$$V(\Delta x) = f[\Delta x, \Omega E(x), M, \text{context}]$$

The model predicts that otherwise comparable changes to strongly Self-linked versus neutral entities should produce different valuation, attention and memory. This is experimentally tractable and separates extensive appropriation from the constitutive body/experience hypothesis.

9.1 Attachment, possession and suffering

Strong attachment can integrate another person deeply into autobiographical prediction without erasing that person's autonomy. MBRS therefore includes Self–Other differentiation as a constraint. Pathological possessiveness is not treated as an inevitable consequence of attachment, but as a possible regime in which extensive belonging, valuation and Self–Other differentiation interact maladaptively.

9.2 Contemplative relevance

The model distinguishes appropriation from attachment. Contemplative practices may reduce recursive narrative elaboration and rigid attachment while leaving minimal bodily reference intact. This provides a testable dissociation: reduced narrative/extended Self gain without abolition of basic body-centred experience.

10. Self–Other modelling and social recursion

$$P_{\text{self}}(t) \neq \hat{P}_{\text{other}}(t)$$

$$\hat{S}_{\text{other}}(t) = H[\text{observation}, S_{\text{self}}(t), \text{memory}, \text{context}]$$

Social cognition requires modelling another agent while preserving referential separation. The model allows simulation of another's state using one's own representational machinery but rejects the idea that action-observation systems alone explain empathy. Social feedback creates an additional recursive loop:

$$\text{SELF_A}(t) \rightarrow \text{OTHER_B}(t) \rightarrow \text{SELF_A}(t+1)$$

Language further allows memories, norms and predictions originating in other people to enter the current informational river. The human Self is thus biologically anchored and socially extended.

11. Computational programme: methods

The computational work accompanying MBRS is exploratory and hypothesis-generating. It was designed to expose circularity, identify dissociations and determine which conclusions follow from explicit assumptions. No simulation is treated as evidence that a machine feels.

11.1 Functional Self without phenomenality

A layered agent architecture included sensors, integration, world-model, organism-reference, own/other discrimination, memory, prediction, valuation, action authorship, appropriation and continuity. Functional Self metrics increased despite the absence of any phenomenal variable.

Functional Self-like organization $\nRightarrow$ feeling

11.2 K experiment: causal invisibility

Three agents received the same sensory stream. A placeholder K was either absent, present but causally disconnected, or coupled to metacognitive report. The disconnected K produced no action/report difference; when coupled, it became part of the functional architecture.

K with no causal consequences $\rightarrow$ empirically invisible

11.3 Regime sweeps

Recurrence, integration, self-reference and persistence were varied. Coordination and persistence increased with coupling, but no non-arbitrary internal threshold licensed the label «consciousness». A subsequent independent four-dimensional sweep exposed an objective-function flaw: high dynamical scores could occur with low self-reference. This motivated replacement of a generic Self score by explicit causal closure and dissociation measures.

11.4 Biofeedback with history

A closed body–representation–valuation–action loop was compared with variants containing weak or strong plastic memory. Closure improved regulation; plastic memory produced path dependence and correlations between prior memory state and subsequent organism representation. The result supports historical dependence, not phenomenality.

11.5 Selective Ω ablation

Ablation	Present-Self proxy	Most affected function	Interpretation within toy model
Baseline	0.867	—	Reference condition
ΩB body	0.375	present reference / Self–Other boundary	Body belonging strongly constrains present reference
ΩX experience	0.525	present reference / boundary	Experiential belonging contributes independently
ΩM memory	0.828	autobiographical continuity = 0.348	Biography can collapse while present reference persists
ΩA authorship	0.704	action authorship = 0.433	Authorship can dissociate from identity
ΩO source	0.787	source monitoring = 0.348	Content and source can

			dissociate
$\Omega B + \Omega X$	0.233	present reference	Combined constitutive perturbation is most disruptive
All Ω	0.034	all outputs	Expected by construction; not evidential

These numerical values are model consequences, not estimates of human function. The scientifically relevant object is the pre-specified pattern of differential impairment. Future computational tests should compare alternative causal architectures under the same observables.

12. Empirical constraints from neuropsychology and experimental neuroscience

The empirical literature is used here as an external constraint on decomposition, not as retrospective proof. The key question is whether real systems exhibit dissociations analogous to those predicted by a multidimensional Ω .

12.1 Body ownership

A 2024 scoping review identified 80 studies including 277 patients with disturbed sense of limb ownership. Tactile and proprioceptive deficits were common, but the review also emphasized incomplete assessment and the role of subjective somatosensory sensations [13]. This supports treating body ownership as a constructed and potentially local relation rather than a unitary body-wide switch.

12.2 Ownership and agency

Systematic and meta-analytic work treats sense of body ownership and sense of agency as distinguishable but interacting components. A 2025 systematic review also emphasized heterogeneity in how agency and body perception are operationalized [14]. MBRS therefore treats ΩB and ΩA as separate dimensions with possible shared mechanisms.

12.3 Source monitoring and hallucinations

A meta-analysis of source-monitoring, self-monitoring and signal-detection studies reported moderate-to-large externalizing effects in hallucination-related samples, particularly in source-monitoring and signal-detection paradigms [15]. A later systematic review/meta-analysis of 44 studies found broad source-monitoring impairment in psychosis and more specific externalizing/imagined-stimulus errors in patients with versus without hallucinations [16]. However, individual studies have also failed to find a general external misattribution bias in first-episode samples [17]. Thus ΩO is a plausible dimension, not a complete theory of hallucinations.

12.4 Dissociative identity disorder

A 2024 systematic review and meta-analysis of inter-identity amnesia included 19 empirical and four case studies. Some meta-analyses supported memory transfer between identity states while others were closer to inter-identity amnesia; the authors highlighted substantial heterogeneity and methodological limitations [18]. A 2024 behavioural/neurobiological study using self-relevant stimuli further illustrates the active empirical debate [19]. MBRS therefore frames DID as a potential test of memory availability versus appropriated accessibility, not as confirmation of a specific mechanism.

12.5 Depersonalization

Recent work links depersonalization experiences to altered self-face representation and interoceptive sensitivity [20]. Because consciousness can remain while the felt relation to Self/body is altered, depersonalization is a high-value target for testing ΩX and bodily reference. The current evidence is not sufficient to reduce depersonalization to appropriation failure.

12.6 Bodily self-consciousness as distributed integration

A 2025 systematic review/meta-analysis of 56 neuroimaging studies found distinct activation patterns for altered body ownership and agency together with shared posterior parietal, precentral and temporo-occipito-parietal involvement [6]. This pattern is compatible with a distributed relational architecture and argues against interpreting any single region as «the Self».

13. Relationship to existing theoretical frameworks

Framework	Overlap with MBRS	Distinct MBRS commitment
Damasio / body-based Self	Organism regulation, body mapping, proto-Self, feeling and levels of Self.	Recurrent appropriation and Γ R are explicit candidate mechanisms for owner emergence.
Bodily self-consciousness	Ownership, self-location, first-person perspective, multisensory integration.	Extends belonging from body to experience, memory, authorship, source and extensive domain.
Global Neuronal Workspace	Distributed non-conscious processing, recurrent amplification, broad availability.	Does not identify global availability with Self; Self is downstream relational organization.
Recurrent Processing	Recurrence as central to conscious processing.	Recurrence is necessary background but not declared sufficient for phenomenality or Self.
Predictive processing / active inference	Prediction, error, action, interoception and organism–world loops.	Adds explicit belonging vector and referential continuity as Self-specific explananda.
Self-model theories	Self as model/process rather than substance.	Claims an ontogenetic causal direction: pre-personal belonging can precede and constitute Self.
IIT	Integration as relevant to conscious organization.	MBRS does not define consciousness by integrated information and makes no Φ -based phenomenal claim.

The model is therefore best treated as a Self-formation framework that can be embedded within more than one theory of consciousness. This modularity is an advantage only if MBRS generates Self-specific predictions that competing frameworks do not already entail.

14. Differential predictions

Prediction	Pre-registered directional claim
P1 — Memory dissociation	Severe autobiographical-memory impairment should reduce autobiographical continuity more than present body-centred self-reference, provided Ω B/ Ω X remain sufficiently intact.
P2 — Agency dissociation	Perturbation of action authorship should be possible with preserved body ownership; the reverse should also occur in at least some paradigms.
P3 — Source dissociation	Source-attribution errors should alter «who/where this content came from» without requiring loss of the content from conscious experience.
P4 — Local body ownership	Local Ω B perturbation should not necessarily collapse global present Self; the magnitude of Self disturbance should depend on hierarchical scope.
P5 — Referential continuity	Across large content changes, measures of Self persistence should track continuity of reference better than similarity of content alone.
P6 — Extensive appropriation	Matched changes to Self-linked versus neutral objects/places should produce stronger valuation, attention and memory proportional to independently measured Ω E.
P7 — Developmental asymmetry	Pre-reflective body/action belonging should emerge developmentally before explicit autobiographical Self and later become bidirectionally modulated by Self-models.
P8 — Contemplative dissociation	Narrative/extended Self gain may decrease during decentring practices while minimal body-centred reference and conscious flow remain.

15. Competing causal models

A scientific test requires alternatives that can fail differently. At minimum, MBRS should be compared with:

Model	Causal core	Critical contrast
MBRS	$\Omega \rightarrow \Gamma R \rightarrow S$, followed by $S \rightarrow \Omega$ recursion	Pre-personal belonging is constitutive.
Self-first model	$S \rightarrow \Omega$	A Self generated elsewhere assigns ownership; Ω is downstream.
Memory-centric model	$M + \text{narrative continuity} \rightarrow S$	Minimal Self should depend more strongly on memory than MBRS predicts.
Predictive-control model	Generative self-model + prediction/action $\rightarrow S$	Belonging adds little once predictive control is modelled.
Integration-only model	S follows sufficiently integrated recurrent state	Selective Ω perturbations should matter only insofar as integration changes.

Future simulations and experiments should compare these models using the same outcome variables and penalize additional parameters. Retrospective weight-tuning after examining clinical profiles should be avoided.

16. Falsification criteria

F1 — CONSTITUTIVE FAILURE

If a stable, integrated present Self is demonstrated while bodily, experiential, causal and temporal belonging relations are systematically absent, the constitutive appropriation hypothesis is strongly weakened.

F2 — DISSOCIATION FAILURE

If selective perturbations of Ω dimensions do not produce preferential changes in their predicted Self functions, the multidimensional decomposition loses explanatory value.

F3 — DIRECTIONALITY FAILURE

If longitudinal or perturbational evidence consistently shows that Self formation precedes all relevant belonging relations and that manipulating pre-personal belonging has no effect on Self development, $\Omega \rightarrow S$ is weakened.

F4 — MODEL DOMINANCE

If a simpler competing model explains the same dissociations, predicts new data better and renders Ω epiphenomenal, MBRS should be revised or abandoned.

F5 — REFERENTIAL CONTINUITY FAILURE

If Self persistence tracks content similarity but not continuity of reference under controlled manipulations, ΓR is not the proposed invariant.

17. Proposed empirical programme

1. Pre-register operational definitions of ΩB , ΩX , ΩM , ΩA , ΩO , ΩE and ΓR before collecting new data.
2. Use factorial virtual-reality paradigms to manipulate body ownership, self-location and agency independently, while measuring present Self continuity.
3. Study neurological amnesia longitudinally with separate measures of episodic memory, autobiographical facts, trait self-knowledge, present self-reference and new learning.
4. Combine source-monitoring tasks with thought/action authorship measures rather than treating hallucination as a unitary endpoint.
5. Develop object/space self-association tasks in which economic value, familiarity and ownership are experimentally separated from ΩE .
6. Use perturbational designs — including sensory conflict, action delays and body illusions — to test causal rather than correlational relationships.
7. Compare MBRS against pre-specified alternative models using held-out data and model-selection criteria.

8. Publish negative and null results, especially cases where appropriation changes without measurable Self changes.

17.1 Measurement strategy

Construct	Candidate observable	Major confound to control
ΩB	ownership ratings, proprioceptive/behavioural indices, multisensory illusion response	demand characteristics, sensory deficit
ΩX	self-presence / experiential belonging ratings, depersonalization measures	general distress, arousal
ΩM	ownership/familiarity of autobiographical material, retrieval profile	memory strength, semantic knowledge
ΩA	intentional binding, explicit authorship, action prediction error	motor performance, task expectation
ΩO	source-monitoring accuracy and bias	recognition memory, guessing
ΓR	cross-state self-continuity measures	content similarity, mood congruence
ΩE	self-linked vs matched-neutral valuation/attention/memory	economic value, familiarity, social desirability

18. Discussion

18.1 What the model currently explains well

MBRS offers a coherent language for separating present self-reference from autobiographical identity, body ownership from action authorship, content from source attribution, and constitutive belonging from later extensive attachment. It also explains why the Self can remain robust under partial damage: Self is not the sum of fixed contents but a relational organization with redundancy and hierarchy.

18.2 The strongest current inference

$$\text{SELF} \neq \sum \Omega_i$$

$$S(t) \approx \text{Invariant}[\Omega(t-\Delta:t), \Gamma R(t)]$$

The strongest theoretical refinement is that Self should not be equated with any individual ownership signal. It is hypothesized to emerge from coherent, temporally persistent relations among such signals. This formulation accommodates partial disownership without requiring disappearance of the entire Self.

18.3 The largest unresolved problem

The direction of causation remains unresolved. Adult Self and appropriation are deeply recursive. Demonstrating that appropriation is constitutive requires developmental, longitudinal or perturbational evidence showing that belonging relations can organize before a mature Self and that their disruption changes Self formation in predicted ways.

18.4 Consciousness remains separate

No result in the present programme licenses the inference that recursive biofeedback, integration, memory or self-reference is sufficient for phenomenal consciousness. This separation is deliberate. MBRS may ultimately interface with GNWT, RPT, predictive processing, IIT or another account, but its current scientific value should stand or fall on Self-specific predictions.

19. Limitations

- The appropriation vector is currently a functional abstraction without a validated measurement model.
- Computational parameters were chosen for conceptual exploration, not fitted to human data.

- Several clinical phenomena used as constraints are heterogeneous and cannot be mapped one-to-one onto Ω dimensions.
- Published evidence may reflect task demands, reporting biases and diagnostic heterogeneity.
- The model risks re-labelling known constructs unless Ω and ΓR generate incremental predictions over ownership, agency, predictive-processing and self-model frameworks.
- The framework currently lacks a quantitative learning rule for the emergence of Ω and ΓR from developmental data.
- The phenomenology of «mine-ness» may not be unitary; bodily, cognitive and social belonging could have partly unrelated implementations.
- No claim of originality is warranted until a systematic review of adjacent theoretical literature is completed.

20. Consolidated MBRS v3.0 hypothesis

FORMAL STATEMENT

A living organism maintains itself through recursive regulation in which bodily state, environmental signals, memory and the consequences of action alter subsequent processing. When part of this dynamics is consciously experienced — by phenomenal conditions not explained here — contents are organized relative to a pre-reflective organism-centred reference. MBRS proposes that recurrent binding of body, experience, memory, action and source to that reference generates referential continuity. The temporally coherent relational invariant that results is the Self. The Self is not an anatomical controller: it is a distributed, virtual and dynamically reconstructed process. Once constituted, it recursively modulates valuation, focus, prediction and action, and can extend belonging relations beyond the biological body.

$$R \rightarrow N \rightarrow G \rightarrow C \rightarrow \Omega \rightarrow \Gamma R \rightarrow S \rightarrow \{V, F, \text{Pred}, A, \Omega E\} \rightarrow R'$$

The arrow $\Omega \rightarrow S$ is the principal constitutive hypothesis and the main point at which MBRS must accept empirical risk.

21. Conclusion

The Recursive Biofeedback Model of the Self reframes the owner problem. Instead of assuming a Self that subsequently owns a body, experiences, memories and actions, it asks whether recurrent relations of belonging can construct the owner. The proposal is compatible with the distributed and dissociable character of bodily self-consciousness, agency, autobiographical memory and source monitoring, but compatibility is not validation.

The model has matured from philosophical intuition to an explicit architecture with defined variables, computational stress tests, competing causal accounts, differential predictions and falsification criteria. Its next scientific step is not additional conceptual elaboration but pre-registered confrontation with data and formal comparison against simpler alternatives.

CORE SENTENCE

The owner need not precede belonging; a stable owner-like Self may emerge from the recurrent continuity of belonging itself.

Appendix A — Computational results retained as constraints

Experiment	Question	Result	What it does NOT show
Functional Self architecture	Can Self-like functions arise without a feeling variable?	Yes, in the toy architecture.	That the agent is conscious.
K experiment	Can a causally inert phenomenal placeholder be detected?	No; disconnected K was behaviourally invisible.	That phenomenality is impossible or epiphenomenal.
Coupled regime sweep	Does more recurrence/integration produce a natural threshold?	No convincing internal critical threshold appeared.	That consciousness has no transition.
4-D phase map	Do dynamical richness scores require self-reference?	No; top regimes could have low self-reference.	That self-reference is irrelevant to real consciousness.
Biofeedback with history	Does plastic memory create path dependence?	Yes; later representations depended on memory history.	That historical dependence equals feeling.
Ω ablation	Can belonging dimensions dissociate?	Yes, by construction, with differential output profiles.	That the same numerical weights hold in humans.

Appendix B — Neuropsychological constraint matrix

Phenomenon	Relevant construct	Observed literature pattern	MBRS status
Disturbed limb ownership	ΩB	Local/limb ownership can be disturbed; somatosensory factors contribute but are not a simple one-to-one explanation.	Compatible constraint
Body ownership vs agency	ΩB vs ΩA	Distinct and overlapping neural/phenomenological components are reported.	Supports decomposition, not directionality
Autobiographical amnesia	ΩM / ΓR	Episodic, factual and trait self-knowledge can dissociate across cases.	Compatible with layered Self
Hallucination/source monitoring	ΩO	Externalizing and internal/imagined source-monitoring biases occur in aggregate, with heterogeneous individual findings.	Partial support for source dimension
DID / inter-identity amnesia	ΩM / ΓR	Subjective amnesia and objective transfer show mixed patterns across studies.	High-value but heterogeneous test
Depersonalization	ΩX / ΩB	Self/body detachment relates to bodily representation and interoception in emerging evidence.	Promising target, underdetermined

Appendix C — Notation

Symbol	Meaning
R	Total internal/external informational flow
N	Distributed non-conscious processing
G	Transient integrated/selected configuration
C	Conscious experiential stream
F	Attentional/focal subset
P	Proto-Self / organism-centred reference
V	Valuation
M	Memory / persistent historical state
Pred	Prediction
A	Organismal action
Ω	Multidimensional belonging/appropriation relation
ΓR	Referential continuity
S	Emergent Self
ΩE	Extensive belonging beyond the biological body

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Reference note

This reference set supports the present scientific white paper but is not a systematic review. A journal submission should replace this section with a documented search strategy, eligibility criteria, risk-of-bias assessment and a structured comparison of competing theories. Claims of MBRS novelty should remain provisional until that review is completed.